

1. Foundations of Sports Data & Context (Weeks 1–4)

Learning Objectives:

- Master global and African sports data ecosystems through direct API work
- Build automated data collection systems from multiple sources
- Create professional-grade data quality assessment tools

Topics to be covered:

- Data Collection and Integration:
 - Api Architecture and usage
 - API connections to Wyscout and Savvy Analytics
 - Web scraping techniques for CAF/FIFA databases
 - Automated data pipeline construction
- Sports Data Architecture:
 - Database design for multi-league systems
 - Cloud storage setup
 - Data versioning and backup protocols
- Quality Control Systems:
 - Missing data detection algorithms
 - Cross-source validation techniques
 - Error correction and flagging systems

Learning Projects:

- Build live data collectors for 3 different leagues
- Create automated quality control dashboard

2. Advanced Sports Analytics Programming (?? WEEKS??)

Learning Objectives:

- Achieve Python proficiency for sports analytics
- Master industry-standard workflows and version control
- Build reusable analytics libraries and tools

Topics to be covered:

- Python Development:
 - Master Sports analytics libraries (mplsoccer, kloppy)
 - Performance optimization for large datasets
 - Master Advance Web scraping techniques
- Industry Workflow Mastery:
 - GitHub version control for analytics projects
 - Docker containerization for reproducible analysis
 - Automated testing and quality assurance
- Advanced Data Processing:
 - Real-time streaming data analysis
 - Multi-format data integration (JSON, XML, CSV, APIs)
 - Parallel processing for performance data

Learning Projects:

- Daily programming challenges with peer review
- Create automated report generation system for quality control flags

3. Data Strategy & Governance (WEEKS??)

Learning Objectives:

- Navigate regulatory environments across regions
- Design data strategies for underserved regions
- Implement privacy-compliant analytics systems

Topics to be covered:

- Strategic Implementation:
 - ROI calculation for data investments
 - Change management for analytics adoption
 - Cross-departmental integration planning
- Regulatory Mastery:
 - GDPR compliance in practice
 - African data protection law navigation
 - Player privacy and consent management
- Organizational Psychology:
 - Resistance to analytics adoption

- Cultural change management
- Stakeholder buy-in strategies

Learning Projects:

- GDPR assessment and improvement plan
- Analytics adoption roadmap with timeline

4. Performance Analytics Mastery (WEEKS???)

Learning Objectives:

- Develop custom metrics and KPIs for different sports contexts
- Master different sports data analysis through hands-on application

Topics to be covered:

- Multi-Modal Data Mastery.
 - GPS tracking analysis
 - Video synchronization with performance metrics
 - Wearable technology integration and interpretation
 - Broadcast data extraction and analysis
- Advanced Metric Development.
 - Custom xG models with validation testing
 - Contextual performance adjustments
 - Opposition-adjusted statistics
 - Weather and venue impact modeling
- Predictive Analytics:
 - Injury risk algorithms with medical input
 - Performance decline detection
 - Load management optimization
 - Recovery protocol effectiveness

Learning Projects:

- Complete GPS tracking analysis system for youth academy
- Use Savvyo Vision Plus for video analysis tool with automated tagging
[you can earn **Savvyo Analytics Certification**: Advanced platform

mastery]

5. Advanced Scouting & Recruitment Analytics (WEEKS??)

Learning Objectives:

- Build comprehensive talent identification systems
- Master cross-cultural player evaluation methods
- Create market inefficiency detection algorithms

Topics to be covered:

- AI-Powered Scouting:
 - Machine learning similarity algorithms
 - Automated highlight video generation
 - Performance trajectory prediction models
 - Cultural adaptation success factors
- Global Market Analysis:
 - Transfer value prediction with multiple variables
 - League strength coefficients
 - Age curve optimization by position
 - Contract negotiation analytics
- Systematic Talent Evaluation::
 - Multi-dimensional player profiling
 - Role-specific evaluation frameworks
 - Team need assessment algorithms
 - Budget optimization models

Learning Projects:

- Transfer decision case studies

6. Business Intelligence & Fan Analytics (WEEKS??)

Learning Objectives:

- Master sports business analytics through revenue data

- Build fan engagement systems
- Create ROI-driven marketing and sponsorship strategies

Topics to be covered:

- Revenue Optimization:
 - Dynamic ticket pricing algorithms with demand prediction
 - Merchandising analytics with seasonal adjustments
 - Concession optimization through fan behavior analysis
 - Sponsorship value quantification with exposure metrics
- Fan Engagement Science:
 - Social media sentiment analysis with monitoring
 - Fan lifetime value calculation
 - Churn prediction and retention strategies
- Digital Marketing Analytics:
 - Multi-channel attribution modeling
 - Influencer impact measurement
 - Brand partnership effectiveness

Learning Projects:

- Revenue optimization dashboard with ROI projections

7. AI & Emerging Technologies

Learning Objectives:

- Build conversational analytics interfaces
- Master computer vision for automated analysis

Topics to be covered:

- Computer Vision Mastery:
 - Player tracking from broadcast video
 - Automated tactical pattern recognition
- Large Language Model Integration:
 - Automated scouting report generation
 - Multi-language analysis
 - Sports knowledge base development

- Predictive AI Systems:
 - Deep learning for performance forecasting
 - Ensemble models for injury prevention
 - Automated decision support systems

Learning Projects:

- Create AI coaching assistant with natural language interface

CAPSTONE PROJECT & INDUSTRY INTEGRATION

CAPSTONE FORMAT:

- Each student works on actual client projects with partner required deliverables

Delivery format

- Biweekly live sessions + asynchronous video content
- Assignments, simulations and peer feedback
- Guest masterclasses

Empieza el curso en marzo.

Estructura y primeros borradores nov-diciembre.

Suelen pagar 80 euros/hora a profes.

Todo en inglés.

Estimamos una **dedicación de 6-8 horas semanales**, donde estarían incluidas las sesiones en vivo.

Por ejemplo, si en la **semana 1** no hay sesiones en directo, la carga no lectiva en casa sería de 6 horas aproximadamente.

En la **semana 2** que sí habría sesiones en vivo, las horas de la sesión (+2h) computan como carga lectiva añadida a esas 6h de carga no lectiva.

Por tanto, habrá semanas con tiempo medio de dedicación 6h y semanas de 8h. En el cómputo total estimamos unas **190h sumando las horas de carga lectiva y no lectiva**.